Project Stochastic Models in Practice
Module 8

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1 Introduction

The traffic flow in hospitals is probably one of the most life-altering applications of queueing theory, since many patients requiring treatment have some form of time limit. Because of this it is of utmost importance for hospitals to have protocols to ensure the relevant patients get their treatment as soon as possible. Most real-life scenarios are, however, already too complicated to analyse mathematically. This is where the use of simulation models can be an alternative for gathering relevant statistical data.

In this report, we will analyse the CT department of a hospital to check if their way of operating ensures a high enough level of safety for emergency patients, and still have enough capacity to scan any other patients in need of it. We will first analyse the situation mathematically, to see if the situation is feasible, and then run a simulation model to gather statistics on safety and flow.

2 Problem description

The hospital has a CT department with two CT-scanners, one of which is only operative during working hours, and a waiting room with three chairs. We are tasked with verifying these are sufficient numbers by checking if people get their scans in time, and how many people have to wait outside of the waiting room because it is full.

The hospital has three different types of patients needing CT-scans, and all three have different ways of arriving to the waiting room for their scan.

Emergency patients. Emergency patients have the most simple arrival pattern, but are also the most time critical to handle. Emergency patients arrive according to a Poisson process with a rate of 24 arrivals per day on average. Emergency patients have direct priority in the waiting room over any in- or outpatients that may already be waiting there.

Inpatients. Inpatients are patients that are already in the hospital than need a CT-scan. An inpatient can request a CT-scan at any time, and they will be called to the CT department whenever the waiting room has no inpatients there and no inpatient is already travelling or there. We will call this request a call from now on since it works similar to outpatients and it is handled similarly in the code. The inpatient then takes a uniformly distributed number of minutes between 9 and 15 to get to the waiting room. Arrivals of inpatients follow a Poisson distribution with a variable arrival rate over time. The arrival rate of inpatient requests is constant outside office hours and follows a sine function with 2 peaks between 9 and 15. The average number of inpatients outside office hours is 6 on a weekday, and inside office hours an additional 21 calls arrive.

Outpatients. Outpatients can call the hospital for an appointment during office hours according to a Poisson process with rate 23 per 8 hours and will be scheduled in 15-minute intervals at the earliest the next day. Before noon there are 4 slots available per hour and after noon 3. Any outpatients that cannot be scanned the same week will be put on a waiting list that gets scheduled into the next week on Friday afternoon.

A CT-scan takes between 10 and 19 minutes for all patient types including any travel required for the scan, so an average of 4.14 scans can happen per hour per CT-scanner

We are tasked with finding the following statistics about the system:

U_{in}, U_{out}	The scanner utilizations inside and outside of office hours.
T_{access}	The average time between an outpatient call and their scheduled appointment.
W_{em}, W_{out}	The average waiting times for emergency patients and outpatients.
O	The fraction of patients that arrive to a full waiting room.
I	The fraction of inpatients with requests in office hours, that did not get scanned before 16:00 the same day.

Table 1: Caption

Our code can be found in the github repository in the folder Assignment3

3 Mathematical analysis

Since we know how many of each patient type will arrive each day and how long a scan takes on average, we can calculate the average scanner utilization inside, and outside of office hours.

Each weekday there is 8 hours of office hours. This totals to 40 hours each week, and the remaining 128 hours are outside office hours. Since each 16 hours outside of office hours 6 inpatients and 16 emergency patients arrive, and the average scan duration is 14.5 minutes the average scanner utilization outside of office hours should be

$$U_{out} = \frac{22 \cdot 14.5}{16 \cdot 60} \approx 0.33.$$

Inside 8 office hours, 21 inpatients, 23 outpatients and 8 emergency patients arrive, and there is twice the amount of scanning time available to avoid congestion. This means the average scanner utilization inside office hours should be

$$U_{in} = \frac{52 \cdot 14.5}{8 \cdot 2 \cdot 60} \approx 0.73.$$

Since scanner utilization is always strictly less than one, the hospital should have enough scanning capacity to scan all patients.

4 Simulation model

For moddeling the system, we use the provided Python simulation and event classes. Building on these, we use 8 different event classes, 2 helper classes and 2 data structures.

The datastructures are `Patient`, keeping track of patient type, arrival time and in the case for out- and inpatient their call time, and `Batch`, keeping track of all statistics within one batch.

The helper classes are `SimInfo` and `Stats`. `Stats` has one `Batch` and has helper functions for updating statistics, starting a new batch and storing the previous one, and calculating all confidence intervals and relative precisions. `SimInfo` keeps track of all other information necessary for the simulation and has helper functions for running the simulation. It keeps track of

- If an inpatient is travelling
- If an inpatient is in the waiting room
- All the inpatients that have called but are not in the waiting room.
- All the outpatients that have called but have no timeslot this week

- All emergency patients in the waiting room
- All in- and outpatients in the waiting room
- The amount of busy scanners
- A list of all scheduled appoinments for outpatients.

and has helper functions for

- Scheduling outpatient appointments throughout the week and on friday for next week.
- Adding patients to the waiting room
- Starting a scan
- Ending a scan
- Calling inpatients to the waiting room

All event classes we use are

- `InpatientRequest` When an inpatient calls
- `InpatientArrival` When an inpatient arrives to the waiting room
- `OutpatientCall` When an outpatient calls
- `EmpatientArrival` When an emergency patient arrives to the waiting room
- `EndScan` When a CT-scanner finishes a scan
- `CheckSchedule` Checks if an outpatient is scheduled and arrives for the current time slot
- `ScheduleOut` Schedules all remaining outpatients every Friday afternoon
- `RefreshBatch` Starts a new batch every batch length

The Simulation initiates according to Algorithm 1.

Algorithm 1 CT-Department simulation initialization

initiate one Simulation, Stats and SimInfo
Schedule the first InpatientRequest, EmpatientArrival, OutpatientCall, CheckSchedule, ScheduleOut, and RefreshBatch
run the simulation according to Algorithm 2

Algorithm 2 CT-Department simulation

while Simulation queue not empty and simulation not terminated **do**
  $E \leftarrow$ next event scheduled
  if E is InpatientRequest **then**
    Schedule next InpatientRequest according to Algorithm 3
    if no inpatient going to or in the waiting room **then**
      Schedule InpatientArrival for the inpatient
    else
      Add patient to the inpatient queue
    end if
  else if E is InpatientArrival **then**
    Add patient to the waiting room, if a scanner is free, start scanning and schedule an EndScan
    log if the waiting room is full
  else if E is OutpatientCall **then**
    Schedule next OutpatientCall, if it is outside office hours schedule it again starting from the beginning of the next working day
    Try to schedule the outpatient this week, if no slot available add the the outpatient queue
    log the difference between call time and appointment time
  else if E is EmpatientArrival **then**
    Schedule next EmpatientArrival
    Add an emergency patient to the waiting room, if a scanner is free, start scanning and schedule an EndScan
    log if the waiting room is full
  else if E is EndScan **then**
    If there is a scanner free and a patient in the waiting room, schedule another Endscan event
    log Scanner utilization
    log waiting time
  else if E is CheckSchedule **then**
    Schedule next CheckSchedule
    if there is an outpatient scheduled for this slot, add them to the waiting room. If there is a free scanner start scanning and schedule an EndScan
    log if the waiting room is full
  else if E is ScheduleOut **then**
    Schedule next ScheduleOut
    Schedule everyone in the outpatient queue for an appointment next week
    log difference between call time and appointment time
  else if E is RefreshBatch **then**
    Schedule next RefreshBatch
    Make a new batch
  end if
end while
Return all statistics

Since inpatients do not arrive according to a Poisson process with a fixed arrival rate, we use the acceptance-rejection method. For this we need the function of the arrival rate of inpatients over time.

Since we know 6 inpatients arrive outside office ours on a workday on average, the baseline arrival rate is $\frac{6}{16}$ per hour. Then inside office hours 21 inpatient calls arrive and the function of arrival rate has a sinusoidal shape with two peaks between 9:00 and 15:00. This gives us a shape for the function:

$$\lambda(t) = \begin{cases} \frac{6}{16} & 9 \geq t \geq 15 \\ \frac{6}{16} + A\frac{1-cos(\frac{2\pi}{3}(t-9))}{2} & \text{Else} \end{cases}$$

We know 21 inpatient calls arrive on average between 8 and 16, so

$$\int_8^{16} \lambda(t)dt = 21.$$

This gives us $A = 6$, so

$$\lambda(t) = \begin{cases} \frac{6}{16} & 9 \geq t \geq 15 \\ \frac{6}{16} + 6\frac{1-cos(\frac{2\pi}{3}(t-9))}{2} & \text{Else} \end{cases},$$

or

$$\lambda(t) = \begin{cases} \frac{6}{16} & 9 \geq t \geq 15 \\ \frac{6}{16} + 6sin^2(\frac{\pi}{3}(t-9)) & \text{Else.} \end{cases}$$

The acceptance-rejection method works by continually drawing a new arrival time T using an arrival rate greater or equal to the maximum arrival rate, and then accepting the time if

$$\text{U(0,1)} \leq \frac{\lambda(T)}{\lambda_{max}}.$$

If it gets rejected, draw a new time starting from the rejected time and try until one is accepted.

Algorithm 3 Schedule next inpatient call

T ← current time + Exp(λ_{max})
while Uni(0,1) $\geq \frac{\lambda(T)}{\lambda_{max}}$ **do**
    T += Exp(λ_{max})
end while
Schedule next inpatient call at time T

5 Verification

5.1 Utilization rates

The realised utilization rates where

$$\overline{U}_{out} = 0.337, \qquad \overline{U}_{in} = 0.658.$$

U_{out} is about 0.004 too high, and U_{in} as about 0.08 too low. This can be explained by in- and outpatients that called during office hours only being scanned after 16:00. There are not that many of them, but the rates are also not very far off their expected values.

Since the total number of available scan hours outside office hours is about two times as high as inside office hours, the contribution to the utilization of any working hour is about two times as low.

5.2 M/M/1 reduction

If we disable all patient types except emergency patients, and limit the number of scanners to 1 at all times and set the scan duration to an exponential distribution, the system reduces to an M/M/1 queue. To check if our simulation functions correctly we can do this reduction and check if the average waiting time is the same as the expected

$$W_q = \frac{\rho}{\mu(1-\rho)}, \qquad \rho = \frac{\lambda}{\mu}.$$

With $\lambda = 1$ emergency patient per hour and an average of $\mu = 4.14$ scans per hour, the expected waiting time is 4.6 minutes.

Running our simulation we get a result of 4.66 minutes with a 95% CI of (4.58, 4.75). So the expected value falls into the CI with a relative precision of 0.1, so we can say that the system was successfully reduced to an M/M/1 queue. This tells us that these parts of the simulation function correctly.

6 Data collection

For collecting the data we use a batch means method, because the simulation does not ever have zero outpatients waiting for an appointment Friday afternoon, so we cannot use this regenerating state for a cycles method.

Since outpatients that get scheduled the next week can influence the amount of outpatients that are left over next week, we set our warm up period to a couple of weeks. The number of outpatients in the queue at the end of the week seemed to settle after 2 weeks consistently, so we chose a warm-up period of 4 weeks, with batches of 16 weeks. We run the simultion for 100 batches and then calculate a 95% confidence interval for all statistics.

We also calculate if the precision is enough for all statistics using the check

$$r \geq \frac{t^2_{r-1,1-\frac{\alpha}{2}} S^2(r)}{(\frac{\delta}{1+\delta}\hat{X})^2}$$

where t is the number of batches and $\delta = 0.1$ is the relative precision

7 Results

Statistic	Mean	95% CI
U_{in}	0.665	(0.663, 0.667)
U_{out}	0.337	(0.336, 0.338)
T_{access}	1.002 days	(0.998, 1.009)
W_{em}	3.120 minutes	(3.088, 3.151)
W_{out}	3.155 minutes	(3.095, 3.215)
O	0.0029	(0.0027, 0.0031)
I	0.031	(0.027, 0.032)

Table 2: Gathered means & CI's

As stated in Verification, Our results line up pretty well with the analytical expected values, and are even more accurate because our analysis of utilization doesn't cover the fact that more inpatients that called during working hours are going to have their appointment after working hours than the other way around.

Most calling outpatients get their appointment scheduled the workday after they call and only have to wait 3 minutes after they show up for their appointment. This short waiting time is slightly shorter for emergency patients, because of their priority. Since their waiting time is less than a fourth of the average scanning time, most emergency patients don't have to wait a significant amount of time before getting their CT-scan.

Only about one in every 500 people finds a full waiting room when they arrive, which is very low. If the hospital wishes to make this near zero, adding just one more chair will drastically lower the already very small percentage of people finding a full waiting room.

About one in every 30 inpatients that called during office hours only get their scan after 16:00, which cannot get much lower, since some inpatients might call the CT department only minutes before office hours end.

Only the scanner utilization outside office hours feels very low at about 33%, but trying to get this number higher would mean either getting slower scanners, which is bad for patients that need a scan as soon as possible, or have more emergency and inpatients arriving outside of office hours. Both of these solutions are clearly much worse than the problem, so the utilization rates look fine.

8 Conclusion

From the results we can see that the hospital is doing very good at this point. Emergency patients experience very short waiting times, and hardly any inpatients are scanned after office hours if they requested a scan within office hours. Furthermore, only 0.3% of all patients encounter a full waiting room.

Based on these results our advice for the hospital is to not buy any more CT-scanners, and at most add a singular extra chair to the waiting room if they wish to reduce the already small probability of patients waiting outside of the waiting room.

9 AI statement

We did not use any AI for making this report or any other part of the project